Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/02/2016
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Project Title:	Coherent Magnon Optics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
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No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mueller, Holger Ph.D. (University of California, Berkeley)		
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The spin degree of freedom of ultracold atoms makes the physics of quantum gases richer, and offers opportunities for fundamental science explorations using cold atomic media. The scientific agenda we propose for the Cold Atom Laboratory (CAL) builds upon on our world-leading research on spinor Bose-Einstein condensates, including our development of magnetization-sensitive imaging and studies of spin coherence in optically trapped quantum gases. We propose four activities:

(1) Precise characterization of magnons in spinor Bose-Einstein condensates: Magnons are low energy spin excitations in spinor condensates. Theories of symmetry breaking, as well as mean-field and beyond-mean-field mean-field treatments of Bose gases provide strong predictions that we will test precisely with interferometric measurements based on coherent magnon optics.

Task Description:	(2) Magnon and Bragg interferometry in dense and dilute Bose condensed gases: magnons in ferromagnetic spinor condensates, such as the $F=1$ gases of 87Rb and 41K, are by necessity gapless and quadratically dispersing, with an effective mass that, within mean field theory that is applicable to weakly interacting gases, is equal to the atomic mass. In other words, magnons propagate just like free particles. We propose to implement an atom interferometer based on coherent magnon optics. Compared to a Bragg interferometer, a magnon interferometer has no mean-field interaction shift. Comparing the two interferometers provides a strong measurement cross-check and may allow quantitative studies of polaronic (quantum depletion) energy shifts. (3) Imaging superflow with a magnon shear interferometer: We propose to image the magnon interference pattern to perform in-situ measurements of superfluid flow in three dimensions, akin to particle tracking velocimetry. We propose to utilize the CAL to expand a trapped Bose-Einstein condensate into a large three dimensional volume, essentially magnifying the superfluid flow pattern sufficiently that it can be imaged. A three-dimensional spatial pattern of magnons will be allowed to propagate coherently before being imaged, revealing the three-dimensional superfluid velocity field. (4) Magnon-evaporative cooling: We will explore a novel cooling technique in which magnon excitations are used to absorb, and then expel entropy from a highly degenerate spinor Bose-Einstein condensate. Our target is to produce quantum gases at the lowest entropy per particle, with a method that can be applied in the CAL apparatus.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is fundamental in flavor. We are exploring new phenomena and new experimental capabilities in a physical system that is itself new and only recently available for experimentation. Exploring in this new space of questions has prospects for making discoveries. More specifically, the research impacts our knowledge of materials (through studies of superfluid turbulence), extends the frontiers of natural phenomena that we can access and explore (through the development of new thermometry and cooling techniques), and contributes to the development of high precision sensors for future scientific and technological applications (through studies of atom and magnon interferometry).
Task Progress:	This narrative describes progress on this project during the year 4/1/2015 - 3/31/2016. We focused on three activities in the previous year: 1) Magnon evaporative cooling and thermometry. Here, we accomplished the goals of Task 4 as listed in our proposal. We found that spin impurities within a Bose-Einstein condensed gas can serve both as a medium for precise thermometry down to very low temperatures, and also as a coolant to reduce the temperature of the trapped gas. In this work, we began with a gas of atoms occupying a single internal atomic state, and cooled down to temperature near or below the Bose-Einstein condensation temperature. We then applied a radio-frequency magnetic pulse to flip the spins of a small fraction of the atoms, creating a minority population of trapped atoms in a different internal atomic state. After allowing the gas to equilibrate at near-constant magnetization, we made state-selective measurements of the atomic momentum distribution. Characterizing the distribution of the minority spin population allowed us to determine the gas temperature down 1 nanokelvin, corresponding to 0.02 times the Bose-Einstein condensation transition temperature. We calculate that, under these conditions, the gas entropy is just S/N = 10^(-3) k_B, meaning that only one out of a thousand atoms in the gas are thermally excited. This is the lowest entropy (quietest) gas ever produced. Previously existing thermometry methods were inadequate to measure such low temperatures/entropies. We observed also that the thermalization of the spin impurities had the effect of cooling the gas. We describe this cooling as an isoenergetic demagnetization cooling effect. Our techniques for thermometry and cooling are compatible with, and have benefits for, the planned CAL instrument. This work is described in Olf et al., Nature Physics 11, 720–723 (2015). 2) Magnon condensation. This activity addresses Task 1 as listed in the proposal (precise characterization of magnons). Magnon condensation describes various equilibrium and non-equilibrium transitions in solid-state materials, in which the population of magnetic excitations in a magnetically ordered system rises above the Bose-Einstein condensation critical number. The ensuing condensation of magnons appears as a spontaneous transverse magnetization, breaking rotational symmetry. We explored this phenomenon within a spinor Bose-Einstein condensate. We introduce a population of magnons into a longitudinally magnetized ferromagnetic spinor condensate. After allowing this magnon population to equilibrate at high temperature, we cool the gas. In this process, the critical number for magnon condensation drops, causing the thermalized magnon cloud to condense within the ferromagnetic gas. We report two main findings. First, magnon condensation occurs as the condensation of free particles in a uniform potential volume defined by the volume of the ferromagnetic condensate. The flat-box effective potential reflects the fundamental property of magnons being gapless excitations. We confirm this description by measuring precisely the position and momentum-space magnon distribution, and also the critical number for magnon condensation. Second, we observe the magnon condensate order parameter directly by imaging the transverse magentization of the gas. We find the order parameter to lack long-range order, with the magnons forming a quasi-condensate. We suspect that the origin of this quasi-condensation is that the Bose-Einstein condensation transition is crossed simultaneously throughout the volume of the ferromagnetic condensate, leading to inhomogeneous symmetry breaking. The dynamics of such symmetry breaking and consequent phase-ordering kinetics, relevant to systems ranging from solids and liquids to the dynamics of the early universe, may be studied in future works. This work is described in our publication Fang et al., PRL 116, 095301 (2016). 3) Quantum gas mixtures of Li and Rb. We are augmenting our experimental capabilities by adding ultracold lithium gases into our experiment. The goal of this activity is to study spinor Bose-Einstein gases with different properties -- e.g., lithium gases in their F=1 state should equilibrate more rapidly than do rubidium gases. We also are interested in preparing gases of ultracold dipolar molecules for applications in precision measurement, quantum dynamics, ultracold chemistry and quantum simulation.
Bibliography Type:	Description: (Last Updated: 06/16/2021)
Articles in Peer-reviewed Journals	Fang F, Olf R, Wu S, Kadau H, Stamper-Kurn DM. "Condensing magnons in a degenerate ferromagnetic spinor Bose gas." Physical Review Letters. 2016 Mar 4;116(9):095301. http://dx.doi.org/10.1103/PhysRevLett.116.095301 ; PubMed PMID: 26991184 , Mar-2016

Articles in Peer-reviewed Journals	Olf R, Fang F, Marti GE, MacRae A, Stamper-Kurn DM. "Thermometry and cooling of a Bose gas to 0.02 times the condensation temperature." Nature Physics. 2015 Sep;11(9):720-3. http://dx.doi.org/10.1038/nphys3408 , Sep-2015
Books/Book Chapters	Stamper-Kurn DM. "Seeing spin dynamics in atomic gases." in "From Atomic to Mesoscale: The Role of Quantum Coherence in Systems of Various Complexities." Ed. S.A. Malinovskaya, I. Novikova. Hackensack, NJ : World Scientific, 2015. p. 61-87. http://dx.doi.org/10.1142/9789814678704_0003 , Jun-2015
Significant Media Coverage	Sanders R. "World's quietest gas lets physicists hear faint quantum effects. News article about PI's research." Berkeley News, August 4, 2015. UC Berkeley online news covering Research, Science, & Environment. http://news.berkeley.edu/2015/08/04/worlds-quietest-gas-lets-physicists-hear-faint-quantum-effects/ , Aug-2015
Significant Media Coverage	Condliffe J. "Scientists Created the World's Quietest Gas to Hear Quantum Effects." Gizmodo, August 5, 2015. Online blurb about PI's research. http://gizmodo.com/scientists-created-the-worlds-quietest-gas-to-hear-quan-1722186207 , Aug-2015
Significant Media Coverage	RT News. "Quantum scientists create 'quietest-ever gas'." RT News, August 6, 2015. https://www.rt.com/news/311734-quantum-quietest-ever-gas/ , Aug-2015
Significant Media Coverage	University of California, Berkeley. "World's Quietest Gas Lets Physicists Hear Faint Quantum Effects." Science Newsline/Physics & Chemistry. Press release by UC Berkeley, August 5, 2015. http://www.sciencenewsline.com/news/20150805100300047.html . , Aug-2015
Significant Media Coverage	Forster F. "Researchers hear noises of quantum effects using quietest gas." Perfscience online media, August 5, 2015. http://perfscience.com/content/2142352-researchers-hear-noises-quantum-effects-using-quietest-gas , Aug-2015